Operating Expenses

	Year Ended December 31,				
				% Change vs. Prior Year	
	2019	2018	2017	2019	2018
		(in thousands)			
Research and development	**$ 87,608**	$ 71,896	$ 57,704	**22%**	25%
Sales and marketing	**56,860**	51,262	47,482	**11%**	8%
General and administrative	**64,603**	52,618	46,054	**23%**	14%
Depreciation and amortization	**7,987**	8,613	9,060	**-7%**	-5%
Restructuring charge	**-**	-	2,921	**NA**	-100%
Operating expenses	**$ 217,058**	$ 184,389	$ 163,221	**18%**	13%

Research and Development

Our principal research and development (R&D) activities during 2019, 2018 and 2017 focused on the expansion and integration of new products and releases, while expanding the product footprint of our software solution suites in Supply Chain, Inventory Optimization and Omnichannel including cloud-based solutions, point-of-sale and tablet retailing.

For 2019, 2018 and 2017, we did not capitalize any R&D costs because the costs incurred following the attainment of technological feasibility for the related software product through the date of general release were insignificant.

Year 2019 compared with year 2018

R&D expenses primarily consist of salaries and other personnel-related costs for personnel involved in our research and development activities. Research and development expenses in 2019 increased by $15.7 million, or 22%, compared to 2018. This increase is primarily due to a $10.6 million increase in compensation and other personnel-related expenses, $3.0 million increase in performance-based compensation expense, and $1.7 million increase in computer infrastructure costs, resulting from increased headcount to support R&D activities.

Year 2018 compared with year 2017

Research and development expenses in 2018 increased by $14.2 million compared to 2017. The increase is primarily attributable to an $8.9 million increase in compensation and other personnel-related expenses, resulting from increased headcount to support R&D activities, a $4.0 million increase in performance-based compensation and a $0.7 million increase in computer infrastructure costs.

Sales and Marketing

Year 2019 compared with year 2018

Sales and marketing expenses include salaries, commissions, travel and other personnel-related costs and the costs of our marketing and alliance programs and related activities. Sales and marketing expenses increased by $5.6 million, or 11%, in 2019 compared to 2018, primarily due to a $4.9 million increase in performance-based compensation expense and a $2.5 million increase in compensation and other personnel-related expenses, offset by a $1.9 million decrease in marketing related expenses.

Year 2018 compared with year 2017

Sales and marketing expenses increased $3.8 million in 2018 compared to 2017, due primarily to an increase of $2.9 million in marketing and campaign programs, a $1.1 million increase in performance-based compensation and a $0.7 million increase in compensation and other personnel-related expenses, partially offset by a $2.0 million decrease in commissions expense as we must defer a portion of our sales commission expense and amortize it over time as the corresponding services are transferred to the customer under ASC 606.

General and Administrative

Year 2019 compared with year 2018

General and administrative expenses consist primarily of salaries and other personnel-related costs of executive, financial, human resources, information technology, and administrative personnel, as well as facilities, legal, insurance, accounting, and other